

SUPPLEMENTARY MATERIAL

Data Processing and Reproducibility Details for the SPARC Rotation-Curve Analysis

Associated study:

PIUM Empirical Galaxy Rotation Curve Re-Analysis: Full SPARC Recalculation from Raw Galaxy Rotation Data

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1. PURPOSE OF THE SUPPLEMENTARY MATERIAL

This document provides the technical details required to reproduce the empirical calculations presented in the study "PIUM Empirical Galaxy Rotation Curve Re-Analysis".

The purpose of the supplementary material is threefold:

1. To describe the exact data-processing pipeline used in the SPARC re-analysis.
2. To provide transparent algorithmic steps for the outer-slope calculation and the Radial Acceleration Relation reconstruction.
3. To allow independent researchers to reproduce the statistical results directly from the publicly available SPARC database.

No processed datasets are distributed here, as all calculations are performed directly from the publicly available observational files.

2. DATA SOURCE

All calculations are based on the SPARC database:

Lelli, McGaugh & Schombert (2016)

SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves.

Public dataset location:

<https://astroweb.case.edu/SPARC/>

The SPARC database provides for each galaxy:

rotation-curve measurements
baryonic mass-model decomposition
global galaxy parameters.

The full catalogue contains 175 disk galaxies.

3. INPUT DATA FILES

The following SPARC files are used:

Rotation curve files
(radial measurements of observed velocity)

Mass model files
(velocity contributions of gas, disk, and bulge)

SPARC master table
(global galaxy parameters including distance, inclination, disk scale length, V_flat and quality flag)

No modification of the observational data is performed.

4. DATA PREPROCESSING

For each galaxy the following quantities are extracted:

radius R
observed velocity V_obs
baryonic velocity V_bar
disk scale length R_disk
quality flag Q
asymptotic velocity V_flat

Only data points satisfying

$V_{\text{obs}} > 0$
 $V_{\text{bar}} > 0$

are retained.

All velocities are converted to SI units (m s^{-1}) and radii to meters before acceleration calculations.

5. OUTER REGION DEFINITION

The outer rotation region is defined using a normalized radius threshold

$R > 2.5 R_{\text{disk}}$

This threshold is selected in order to minimize contamination from the rising inner disk region.

Galaxies are accepted for the outer-slope analysis only if

quality flag $Q \leq 2$
at least 8 radial measurements satisfy the outer-region condition.

6. OUTER SLOPE CALCULATION

For each accepted galaxy the outer rotation curve is fitted using a log–log linear regression.

The fitted relation is

$\log_{10}(V_{\text{obs}}) = a + s \log_{10}(R)$

where the slope

$s = d \log_{10}(V_{\text{obs}}) / d \log_{10}(R)$

represents the outer rotation-curve slope.

Regression is performed using ordinary least squares (OLS).
No weighting is applied in the baseline analysis.

7. OUTER SLOPE STATISTICS

The ensemble statistics reported in the main paper are computed across the full accepted galaxy sample.

For the fiducial outer definition

$R > 2.5 R_{\text{disk}}$

the accepted sample contains

79 galaxies.

The slope distribution statistics reported in the main paper are derived from this sample.

Detailed statistical values (mean slope, median slope, dispersion and range) are presented in the main article.

8. RADIAL ACCELERATION RELATION CONSTRUCTION

For each radial measurement two accelerations are computed.

Observed acceleration:

$g_{\text{obs}} = V_{\text{obs}}^2 / R$

Baryonic acceleration:

$g_{\text{bar}} = V_{\text{bar}}^2 / R$

Both quantities are expressed in SI units (m s^{-2}).

All radial measurements with valid velocities are included.

9. GLOBAL RAR RELATION

The global Radial Acceleration Relation is constructed by plotting

$\log(g_{\text{bar}})$ versus $\log(g_{\text{obs}})$

for all accepted radial measurements.

Correlation properties reported in the main study are obtained by linear regression in log–log space.

10. SLIDING WINDOW SLOPE ANALYSIS

To examine the internal structure of the RAR, a sliding-window analysis is performed along the $\log(g_{\text{bar}})$ axis.

Parameters used:

window width = 0.3 dex
step size = 0.05 dex
minimum points per window = 50

Within each window a local linear regression is computed.

The resulting slopes characterize the gradual transition between dynamical regimes.

11. EMPIRICAL RAR APPROXIMATION

The RAR transition curve can be approximated by the empirical expression

$g_{\text{obs}} = g_{\text{bar}} / [1 - \exp(-\sqrt{(g_{\text{bar}} / g_0)})]$

where

$g_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$.

This expression is used purely as an empirical interpolation of the reconstructed RAR and does not represent a new dynamical law.

12. REPRODUCIBILITY

All calculations described above can be implemented using standard numerical tools such as Python with NumPy and SciPy.

The analysis requires only the publicly available SPARC files and does not depend on proprietary data.

No parameter tuning or galaxy-by-galaxy adjustments were applied in the baseline calculation.

Independent researchers can reproduce the full set of results by applying the algorithmic steps described in this document.

13. FUTURE WORK

The empirical results described in the main paper suggest a structured transition between dynamical regimes in disk galaxies.

Future work may test whether this observed radial transition can be reproduced in simulated galaxy populations generated under different dynamical prescriptions, including dark-matter halo models, modified gravity frameworks, or collective field-based descriptions.

END OF SUPPLEMENTARY MATERIAL